Urban Transit Infrastructure and Inequality*

Kwok Hao Lee† Brandon Joel Tan‡

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Abstract

We propose a quantitative spatial model featuring heterogeneous worker groups and travel to consume non-tradable goods and services. Using transit farecard data from Singapore, we establish that low-income workers are more sensitive to changes in travel time than high-income workers. Since the former are overwhelmingly employed in non-tradable sectors, changes in consumption travel can induce a spatial re-organization of low-income jobs in the city. The Downtown Line resulted in large welfare gains for high-income workers, but near zero for low-income workers. All workers enjoyed improved access to consumption opportunities, but low-income jobs in the non-tradable sector moved to less attractive workplaces. Abstracting from consumption travel understates the disparate impact of the DTL across worker groups three-fold.

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†Corresponding Author; Department of Strategy and Policy, National University of Singapore Business School, kwokhao@nus.edu.sg

‡Department of Economics, Harvard University, btan@g.harvard.edu

1 Introduction

With 68% of the world's population projected to live in urban areas by 2050, governments will continue to spend vast sums on expanding mass transit (United Nations 2018). As with any large public investment, it is important to consider how the economic benefits are shared between low- and high-income workers. If the economic benefits of transit are not shared, urban inequality may be exacerbated.[1] Urban inequality undermines social cohesion and may entrench accumulated advantages, thus threatening intergenerational mobility (see, e.g., Chetty et al. 2014).[2]

Existing research has highlighted differential access to employment opportunities across worker groups, emphasizing that poor workers face much larger commuting costs than the better-off. However, a substantial portion of transit trips are unrelated to work. These trips are taken for the consumption of non-tradable goods and services, such as visiting restaurants, retail stores, and salons.[3] Since low-income workers are overwhelmingly employed in non-tradable sectors (such as waiters, retail workers, manicurists and so on), changes in consumption travel lead to a spatial re-organization of low-income jobs in the city, possibly leading to disparate impacts of transit expansion by income stratum.[4]

In this paper, we ask the question: do the poor and other workers benefit alike from transit expansion, and what role does consumption travel play in this analysis? We exploit the universe of transit farecard trips from Singapore during the opening of the Downtown Line (DTL), as described in Section 2. These farecard data are combined with administrative spatial data on land

[1] Public transit systems can disproportionately serve higher income workers. For instance, McKenzie (2015) shows that higher income workers are more likely to live in public transit accessible neighborhoods in Washington D.C..

[2] In Singapore, real household income for the bottom 10% increased 46.2% between 2000 to 2022, while it increased 90.5% and 77.9% for the top two deciles respectively (Singapore Department of Statistics). In the United States, the Gini coefficient for inequality has increased from 0.36 to 0.43 for urban counties between 1970 and 2016 (Thiede et al. 2020).

[3] According to the United States National Household Travel Survey, as of 2017, 31% of travel miles are made for work purposes, while 33% of travel miles are made for shopping or meals, and another 26% for social or recreational purposes. See Appendix Figure A2 (Department of Transportation 2017). Additionally, in 2018, non-tradable consumption accounted for over 50% of total household expenditure in Singapore (Department of Statistics 2019). See Appendix Figure A3.

[4] Disparate impacts, in and of themselves, do not mean that transit expansion is a policy governments should not pursue, especially if no one is strictly worse off. After all, high-income workers are net payers into most social support systems. Nonetheless, many governments aim to reduce social disparities through investments in transit.

use, rents, expenditures and employment. The data inform a novel quantitative spatial model with heterogeneous worker types and travel to consume non-tradable goods. Our goal is to shed light on the interplay between three pieces of the puzzle: consumption travel, inequality, and the spatial re-organization of economic activity.

The core messages of our paper are threefold. First, using gravity regressions, we find that low-income workers are more sensitive to changes in travel time than their high-income counterparts.[5] Second, we establish that low- and high-income workers live, work, and consume non-tradables in different places. Thus, the placement of the DTL impacts each worker type differently. Third, since low-income workers are overwhelmingly employed in non-tradable sectors, we find that changes in consumption travel induced a spatial re-organization of low-income jobs in the city, with important distributional implications.

Constructed at a cost of US$15.5 billion, the DTL is the longest underground mass rapid transit line in Singapore, and is the city-state's most ambitious public transit project to date (The Straits Times 2017).[6] In Section 3, we present evidence that, when a section of the DTL opened in December 2015, low- and high-income workers adjusted their travel patterns differently in both the short run and up to a year after the line opened. Additionally, shortly after the precise locations of DTL stations were announced in 2008, we find that the announcement coincided with higher price appreciation for apartments near each DTL station, compared to those farther away.

Motivated by the descriptive evidence, in Subsections 5.1 and 5.2, we use gravity regressions to assess how responsive each type of worker is to travel time. We find that low-income workers have larger travel elasticities than high-income workers, and elasticities are larger for consumption than for work travel across both types. These findings are robust to a differences-in-differences analysis comparing origin-destination pairs treated by the line opening (Severen 2023), and accounting for sparsity in the matrix of said origin-destination pairs (Dingel and Tintelnot 2020). Prior to the opening of the DTL, low-income workers commuted shorter distances than high-income workers

[5] In Singapore, low-income workers earn a monthly salary below the 25th percentile, and are targeted for transit subsidies and income support.

[6] The DTL sought to provide direct access to the city center from the north-west and east. As a "closed city" with restrictions on immigration, any welfare gain from improved transit infrastructure is captured by Singaporeans.

do, but traveled as far as the latter to consume non-tradables. While our gravity regressions provide a framework to understand the data, more structure is needed to assess how the DTL impacts welfare and inequality in Singapore.

To interpret these empirical findings, we develop a quantitative spatial model in Section 4 that nests several others. In addition to incorporating heterogeneous worker types by income (low/high), our main theoretical contribution is modeling travel to consume non-tradable goods in a manner that is informed by the data and admits tractable counterfactuals using exact-hat algebra (Dekle, Eaton, and Kortum 2008). In the model, workers travel to work and to consume non-tradable goods and services. The city in which they reside is modeled as a set of neighborhoods, with a transportation network characterized by bilateral travel times between locations. Worker groups differ in their preferences over residences and consumption locations, productivities over workplaces and sectors, consumption patterns, and travel costs and elasticities. Workers first choose where to live, then where to work, and finally where to consume non-tradables.

When choosing where to live, each worker trades off rents against residential amenities, as well as access to employment and consumption opportunities. When choosing where to work, each worker evaluates their type-specific wages, match-specific productivities and commute distances. Finally, when choosing where to consume non-tradables, they consider prices, type-specific idiosyncratic consumption amenities, and travel distances. On the production side, the non-tradable and tradable sectors have different input requirements over commercial floor space and labor provided by each worker group, with the non-traded sector being much more intensive in low-income labor. Market clearing in the non-traded sector implies that, in each location, consumption travel patterns drive the demand for non-traded production, and hence also the demand for low-income workers. Finally, the model also accommodates agglomeration forces, and allows for type-specific spillovers in endogenous residential amenities. These spillovers drive segregation in the city.

With the model in hand, to undertake our counterfactuals, we estimate and calibrate several other key parameters in Section 5. Through the lens of the model, the fixed effects in our gravity regressions can be interpreted as measures of location attractiveness. These measures are validated

against external data on associated amenities. Model-implied average wages by worker type and residence are also highly correlated with those observed in administrative data. Finally, using changes in travel times as observed in the farecard data before and after the opening of the line, we find that the model predicts well post-DTL travel flows, residential patterns and the entry of firms in the non-tradable sector.

After having computed our estimates, using exact-hat algebra, we quantify the welfare and inequality effects of the DTL in Section 6. We find that the DTL improves welfare for high-income workers by 1.8%. However, low income workers experience almost no net benefits.[7] Although both groups can better access consumption opportunities, low-income workers see their jobs in the non-tradable sector moving to places they deem less attractive, offsetting their gains from the consumption channel. This disparity between high- and low-income workers is driven by two mechanisms. First, the DTL disproportionately improves access for many high-income areas served directly by the line. Second, more consumers consume non-tradables near the city center because the DTL has made it easier for them to go downtown. Thus, low-income jobs in the non-tradable sector move to the city center, yet low-income workers still live away from the center. Average commute times increased about 1% for low income workers, while high income workers saw a reduction.

Abstracting away from travel to consume non-tradables, we find a three-fold underestimation of the disparate impact of the DTL across worker groups. We find that both worker groups benefit from greater access to employment opportunities but a slightly larger share of the gains go to high-income workers. Absent non-tradable consumption, the model misses the spatial re-organization of low-income non-tradable sector jobs. Additionally, the model ignores the fact that workers value access to consumption opportunities, understating the aggregate welfare effects of the DTL, and failing to capture the disproportionate gains in consumption access for high-income workers.[8]

[7] In Subsection 6.3, we find that these implications for inequality are robust to equalizing, between low- and high-income workers, expenditure shares; travel costs; travel elasticities; and residential elasticities.

[8] Additionally, we consider the role of line placement in driving our results. In Online Appendix Section F.2, we simulate the removal of the North-South Line (NSL), a trunk line primarily serving more diverse neighborhoods. When the NSL is removed, welfare falls for both low and high types, by 4.2% and 2.0% respectively. Low income workers experience a 2.2 p.p. greater loss in welfare compared to high income workers. Both types experience reductions

This paper speaks directly to the literature studying the impact of transportation infrastructure within cities (McDonald et al. 1995; Gibbons and Machin 2005; Baum-Snow and Kahn 2005; Glaeser, Kahn, and Rappaport 2008; Billings 2011; Donaldson and Hornbeck 2016; Severen 2023; Donaldson 2018; Brooks and Lutz 2019; Gupta, Van Nieuwerburgh, and Kontokosta 2020; Heblich, S. Redding, and D. Sturm 2020). The most closely related papers to ours are Tsivanidis (2019) and Balboni et al. (2020), who study the Bus Rapid Transit (BRT) systems focusing on inequality in Colombia and Tanzania respectively. Both papers estimate quantitative spatial models restricted to workplace commuting, finding small inequality effects. We innovate on the literature by incorporating the role of travel to access non-tradable goods and services. These travel costs are crucial to evaluate the distributional welfare effects of transit expansions. Consistent with prior work, when we abstract from consumption trips, we find the DTL barely has disparate impacts by income. However, factoring in consumption trips gives us the opposite conclusion.

Our paper also contributes to several other strands of literature. The first is the growing body of work on quantitative spatial models more generally (S. Redding and Daniel M. Sturm 2008; Allen and Arkolakis 2014; Ahlfeldt et al. 2015; S. Redding 2016; Allen, Arkolakis, and Li 2016; Caliendo, Parro, et al. 2018; Desmet, Nagy, and Rossi-Hansberg 2018; Monte, S. J. Redding, and Rossi-Hansberg 2018; Adao, Arkolakis, and Esposito 2019; Allen and Arkolakis 2022).[9] Our paper is closely related to that of Miyauchi, Nakajima, and S. Redding (2022), who explore the role of trip chaining in explaining consumption externalities across locations in Tokyo. Unlike theirs, we are the first to explore inequality in work and consumption access, in a parsimonious yet tractable manner, through a quantitative spatial model.

This paper also speaks to quantifying the value of urban amenities (Roback 1982; Blomquist et al. 1988; Glaeser, Kolko, et al. 2001; Albouy et al. 2016; Cragg and Kahn 1997; Bayer et al. 2009, Fan, Guthrie, and Levinson 2016; Diamond 2016; S. J. Redding and Daniel M Sturm 2016;

in consumption and employment access. Thus, line placement matters when considering the implications of transit expansion on welfare and equity.

[9] A growing strand of this literature focuses on dynamic effects (see, e.g., Kleinman, Liu, and S. J. Redding 2023). Our paper focuses on the comparison between the pre- and post-Downtown Line steady states; thus, a static model suffices. The question of where to jointly locate housing and transit, as well as dynamic effects, is left for future work.

Almagro and Domínguez-Iino 2020). We demonstrate the important role of non-tradable retail amenities in shaping the spatial distribution of economic activity in the city. Moreover, we are also able to quantify differences in preferences over amenities across worker groups. Finally, our paper is related to the literature on inequality in cities (Brueckner et al. 1999, Glaeser, Resseger, and Tobio 2009; Su et al. 2018; Fogli and Guerrieri 2019; Couture et al. 2019). We show that changes in travel access can have substantial disparate impacts by income group.

2 Background and Data

2.1 Context: Singapore and the Downtown Line

Singapore is an island city-state in Southeast Asia. With a population of about 6 million inhabitants, Singapore is the third densest country in the world, and is among the densest cities (World Bank 2019).[10][11] Inequality in the city-state is high; with a Gini coefficient of 37.8 as of 2022, Singapore ranks between the United States and Chile among high-income countries.[12] Of late, inequality in the city-state has been of concern because the rise of conspicuous consumption has eroded the social fabric (Y. Y. Teo 2022). When pressed to redistribute via higher taxes and social spending, the government has demurred, pointing to the need to maintain the city's attractiveness as a place to do business.[13]

The population in Singapore is heavily reliant on public transportation, comprising buses, light-rail and mass-rail networks. The Singapore government heavily restricts the supply of cars with a Vehicle Quota System. As of 2018, the price of a Volkswagen Golf 1.4 is US $110,479.80, 5 times more expensive than in the United States (US$21,845). As a result, a minority of households in Singapore own a car, and far fewer drive to work.[14] With the public transit system carrying

[10]There are highly restrictive immigration policies with strict foreign worker quotas, and limited out- and in-migration (Ministry of Manpower 2020).

[11]The spatial distribution of non-tradables across Singapore is similar to that of other major cities (see Figure A4).

[12]This Gini coefficient is based on household income from work per household member, after adjusting for government taxes and transfers. See `https://www.singstat.gov.sg/-/media/files/news/press09022023.ashx` for details.

[13]Academic and popular discourse in Singapore highlights a culture of "neoliberal morality" (e.g., Y. Teo 2013). In Singaporean society, as opposed to the broad-based welfare system in Denmark or Norway, income and consumption taxes are kept low, and the family is elevated as the main social safety net. Where family support falters, the poor can turn to government aid, but applying for welfare can be onerous due to involved paperwork and rigorous means-testing.

[14]As of November 2022, only 471,000 households of 1.7 million (28%) own a car; 56,250 own two; and 14,130

over 4 million passengers per day, public transportation is the primary means of travel for the population.[15] Approximately 60% of trips are made on buses, as opposed to rail lines.

The Downtown Line is the longest underground and mass rapid transit line in Singapore. At 41.9 kilometres (26.0 miles) and with 34 operation stations, the line runs from Bukit Panjang station in the north-west to Expo station in the east via the Central Area.[16] The line was first announced on 23 October 2001 and was built in three phases.[17] The first phase opened in December 2013, and was composed of 6 stations from Bugis to Chinatown station within the Downtown area. The second phase from Bukit Panjang to Rochor station, which linked the north-west to the center of the city, opened in December 2015. The third and final phase from Fort Canning to Expo station, connecting the east to the city of the city, opened in October 2017. The line, at a cost of 15.5 billion USD, is considered the government's most ambitious public transit project to date (The Straits Times 2017). The purpose of the line was to provide the north-west and eastern areas a direct link to the center of the city, and to alleviate congestion on various other rail and bus routes.

2.2 Data

Our primary source of data is public transit fare card data (EZ-Link) from the Land Transport Authority of Singapore.[18] We observe all trips made by public transit (mass rail, light rail or bus) linked to an individual's "EZ-Link" card. Our data set covers one week every quarter between June 2015 and June 2018, and three full months between December 2015 and February 2016. The longer quarterly data set allows us to observe changes in transit patterns from before the opening of Phase 2 of the Downtown Line to after the opening of Phase 3. The full three months of data

own three cars or more.

[15] Appendix Table A1 provides a summary of the mass-rail lines in Singapore.

[16] See Figure A5.

[17] At the opening of Dover Station on the East-West Line, the then Minister for Communications and Information Technology Yeo Cheow Tong announced that Stage 2 of the Downtown Line "will serve the Bukit Panjang, Upper Bukit Timah and Bukit Timah regions... and will alleviate the heavy traffic in the Upper Bukit Timah/Bukit Timah/Dunearn Road corridor." His speech did not contain further information on line alignment or where stations would be sited. On 15 July 2008, in a press release by the Land Transport Authority of Singapore, the precise location of stations on the line was revealed.

[18] A heat map visualization of the data is presented in Appendix Figure A7. For data construction and auxiliary data sets, see Online Appendix Section A.

captures the period directly before and after the opening of Phase 2 of the Downtown Line. In total, we observe over a billion trips. For each trip, we observe the origin, destination, and start and end time of the trip. Each individual in our data set is categorized into commuter groups. In our analysis, we focus on the Adult and Low-Income Worker categories. Low-Income Workers are those who earn a monthly salary below the 25th percentile (S$2,000 prior to 2020, against a median of S$4,534).[19]

Our unit of analysis is the *subzone*, the smallest administrative planning unit in our data sets, delineated by the Urban Redevelopment Authority of Singapore. Singapore is divided into 323 subzones with a median size of 1,229,894 square meters. These subzones are contained within larger spatial units, namely 55 planning areas and 5 planning regions. We link all bus and train stops to subzones.

We use our fare card data to generate work and consumption travel probabilities conditional on residential subzone. We identify each individual's residence as the modal first origin and last destination of the day. We identify each individual's workplace as the modal destination during the morning rush hour (5am to 11 am) and origin during evening rush hour (3pm to 11pm).[20] Finally, we classify all remaining trips as consumption trips.

We also use data from several other administrative sources and spatial data on amenities from the government data portal (Open Government Products 2019). See Online Appendix Section A for details on these data sets and their construction.

3 Descriptive Analysis

In this section, we present descriptive results on the impact of the Downtown Line and motivating facts that guide our empirical analysis. In all our analysis, we define low-income workers as those earning below the 25th percentile ($2,000 SGD or $1,500 USD per month, less than half of median earnings). All others are high-income workers, consistent with our farecard data. Our unit

[19]Low Income Workers receive up to a 25% subsidy on their fare costs. The Central Provident Fund automatically determines eligibility based on tax filings and sends an individual application package informing workers of their eligibility, details of the scheme, and how to apply for the subsidized card (Ministry of Transport 2014).

[20]Modal morning destination/evening origin locations match 71% of the time and we validate that these identified residences and workplaces match well their corresponding shares in administrative data (Figure A15). See Online Appendix Section A for details.

of analysis is the subzone, which we will henceforth call a neighborhood.

3.1 Residential, Employment, Consumption, and Travel Patterns

First, low- and high-income groups have different residential patterns across the city. Figure 1 plots the share of high-income workers that live in each neighborhood in 2015. High-income workers primarily cluster near the center of the city (e.g., Bukit Timah, Tanglin) with only some concentration in certain neighborhoods near the coasts of the island (e.g., East Coast Park). On the other hand, low income workers live farther from the city center, with many living closer to the north. Figure 1 also shows that the Downtown Line runs through some of the neighborhoods with the highest concentration of high-income workers in Singapore, clustered in or just west of the city center. The many low income residents in the north are not directly served by the Downtown Line. This suggests that high-income workers may disproportionately benefit from the DTL, since the line serves them directly.

Second, neighborhoods with a high concentration of low-income workers are associated with a high share of non-tradable sector production. Figure A11 shows a strong positive correlation between the share of commercial land used by the non-tradable sector and the share of workers that are low-income. This finding is in line with the Labor Force Survey data, which indicates that 51% of workers in the non-tradable sector are low income, compared to only 14% in the tradable sector, as shown in Table A2.[21] These findings highlight the importance of the spatial distribution of non-tradable jobs for low-income workers.

Third, using itemized Household Expenditure Survey data broken down by income group, we find that high-income workers spend more on non-tradable goods and services than low-income workers. Figure A12 shows that high-income workers spend 40% of their income on non-tradable goods and services, 41% on tradables, and the remaining on housing. In contrast, low-income workers spend 34% of their income on non-tradable goods and services, 43% on tradables, and the remaining on housing. These findings suggest that improving access to consumption opportunities

[21] Specifically, 61% of employees in the food and accommodation, 49% in retail, and 41% in personal services are low-income. Low-income workers only comprise 25% of the labor force in Singapore.

will raise welfare more for high-income than low-income workers, as the former spend a larger share of their income on non-tradables.

Fourth, low income workers make shorter commutes, at a median of 25 minutes, compared to 30 minutes for high income workers. However, for consumption trips, travel times are more similar across income groups and are shorter than workplace trips, at around 23 minutes for high-income workers and 22 minutes for low-income workers respectively. Figure A8 plots the travel time distributions by worker group and type of travel. Low-income workers make 5% more trips on average than high-income workers. Consumers also make more trips on the weekend than weekdays. Figure A13 presents the average number of daily trips by worker group and weekday vs weekend travel.

3.2 The Downtown Line and Housing Prices

The precise locations of stations on the Downtown Line (DTL) were suddenly announced on July 15, 2008. We show that the announcement coincided with large changes in residential prices for houses near each DTL station. Our empirical design is an event study across time and space.[22] We compare "treated" apartments within one kilometer of a DTL station, relative to "control" apartments between one and five kilometers from one.[23] "Treated" apartments are more likely (than "control" apartments) to directly benefit from access by foot to a DTL station, in addition to the entry of non-tradable services in the vicinity of the station.[24]

Over 4 years, relative to apartments in the outer ring, residential prices increase by 4.84% for apartments in the inner ring. See Figure 2a.[25] Moreover, we also estimate the relationship, over

[22]For details on the precise specification and analysis, see Online Appendix B.1.

[23]Our results are robust to shrinking the outer limit from 5km (2-5km being the control group) to 4km (2-4km being the control group) and 3km (1-3km being the control group). See Table A5.

[24]To interpret our subsequent results as causal (a difference-in-differences design), one would have to believe that prices of treated and control apartments would have evolved similarly in the absence of the DTL. Any confounding factor driving housing prices must differentially affect apartments close to and farther from a DTL station. While an inspection of Figure 2a suggests that the parallel trends assumption holds, a large section of the DTL was built to relieve congestion along Bukit Timah Road and Dunearn Road, along which the line runs. Moreover, control apartments may also be affected by existing stations on other lines receiving different amounts of traffic because of the opening of DTL Phase 2.

[25]Similarly, apartments for which the distance to a Phase 2 DTL station was less than the expected distance to a station (assuming that stations are random uniformly distributed along the DTL) increased in prices after the an-

time, between housing prices and distance from the DTL. The estimated regression coefficients are plotted in Figure 2b. Post-announcement, the closer the apartments are to the DTL, the higher their prices, all else equal.

Increased property prices suggest that residents highly value improvements in access from the DTL.[26] These price increases could reflect both improved access to work and consumption locations. We find that these price effects increase over time, suggesting a longer-run re-organization of economic activity.

3.3 Impact of the Downtown Line on Travel

When Stage 2 of the Downtown Line (DTL) opened, we saw a sharp uptick in travel to DTL subzones. We conduct an event study around the opening of Phase 2 of the DTL (DTL2), which took place on December 27th, 2015. The immediate pre-period, from December 1 to 26, forms the relevant short-run counterfactual because any disruptive road works would have been completed months before the opening of the line.[27] We compare daily trips taken, by income type, to DTL and non-DTL subzones, before and after DTL2 opened.

Figure 3a plots the main results. We observe about a 20% increase in trips after DTL2 opened. Over all subzones, the volume of low-income trips increases by a smaller percentage than high-income trips. This finding suggests that high-income workers are taking greater advantage of the new DTL than low-income workers.

We also find that low- and high-income workers adjust their travel patterns differently across neighborhoods. In Figure 3b and 3c, we plot the travel patterns by income group for two of the major locations on the DTL line separately, Bukit Panjang and Upper Bukit Timah. High-

nouncement relative to apartments where the distance to a Phase 2 DTL station was more than the expected distance. See Appendix Section B.1, Table A5, and Figure A14).

[26] We note that much of the housing stock (80%) in Singapore is government housing (see, e.g., Wong 2013, Ferdowsian, Lee, and Yap 2023 and Lee, Ferdowsian, and Yap 2023), bought and sold on long leases. Regulation in this market is intentionally kept light-touch to facilitate price discovery and asset appreciation for existing owner-occupiers (see, e.g., Chua 2017).

[27] Much of the time immediately preceding the opening of train lines is spent on train testing. For instance, Phase 3 of the DTL opened in October 2017, and was tested extensively by its operator SBS Transit from May 2017 to August 2017. See the following link for details:
`https://www.todayonline.com/singapore/downtown-line-start-later-every-sunday-may-august`

income workers increase their travel to and from Bukit Panjang by about 40% on average, while low-income workers increase their travel volume by about 20%. On the other hand, both worker groups increase their travel to and from Upper Bukit Timah by the same amount. This disparity suggests that our model should account for differential travel responses when evaluating welfare across worker groups.

Lastly, we plot changes in travel patterns over a longer time horizon in Figure A16. We find that the responses in travel flow are grow over time, stabilizing about a year after the line opened. Therefore, it seems important for us to account for the long-run re-organization of economic activity in Singapore.[28]

4 Quantitative Model

To guide our empirical analysis, we model a closed city with heterogenous workers and travel choice over workplace and consumption locations. The city has $n \in \mathbb{N}$ neighborhoods.[29] Neighborhoods differ in their exogenous amenities, productivities, residential and commercial floor space, as well as bilateral commute times.

High- and low-income workers decide where to live, where to consume non-tradable goods and services, and where and in which sector to work.[30] Each worker type has different preferences, productivities, wages, travel costs, and consumption shares. Utility is derived from the consumption of a tradable good, a non-tradable good, and residential floor space.

In the production side of the model, there are two sectors: tradable goods (T) and non-tradable goods (N). Firms in both sectors are located across the city and use labor and commercial floor space as inputs. The demand for different worker types differs across sectors, with the non-tradable sector relying more on low-income workers, while the tradable sector relies more on high-income

[28] Additionally, using two cross-sections of food establishment license data between 2015 (before the opening of Phase 2 and 3 of the DTL) and 2018 (after the opening), we find that, in subzones with a DTL station, relatively more food establishments entered between 2015 and 2018. The number of food establishments in DTL locations increased by about 15% relative to non-DTL locations. See Appendix Table A4.

[29] The population of each worker group is determined exogenously. Singapore is a small city-state with tightly regulated in- and out-migration.

[30] In additional analyses, we allow each worker to make many consumption trips to different locations. We find that this extension collapses to the same estimating equations as in the main model. This material is available in a supplement provided by the authors upon request.

workers. The demand for non-tradable goods depends on where consumers choose to travel to consume them, while tradable goods can be costlessly traded across the city. The demand for labor by worker type varies across the city, depending on the productivity of each sector in each location, commercial rents, and demand for production. Perfectly competitive developers supply floor space using land and capital with constant returns to scale technology. In equilibrium, wages and the prices of floor space and goods adjust to clear markets in labor, land, and goods in the city.

4.1 Workers

The city is populated by different worker groups indexed by $\theta \in \{+,-\}$, denoting high- and low-income workers respectively. Each type of worker has a fixed population $R(\theta)$.[31] A worker ω in group θ chooses a location n in which to live, a location i and sector G in which to work, and a location j in which to consume non-tradable goods and services. We assume indirect utility is Cobb-Douglas and takes the form:

$$u_{nij}(\omega;G,\theta) = \frac{b_n(\omega;\theta)}{Q_n^{\gamma(\theta)} P_n(T)^{1-\alpha(\theta)-\gamma(\theta)}} \times \frac{w_i(G,\theta)a_i(\omega;G,\theta)}{\exp(\kappa(\theta)\tau_{ni})} \times \frac{s_j(\omega;\theta)}{P_j(N)^{\alpha(\theta)}\exp(\kappa(\theta)\tau_{nj})}, \quad (1)$$

where $0 < \alpha(\theta), \gamma(\theta) < 1$. To ease exposition in this subsection, fix a residential neighborhood n, work neighborhood i, sector G, consumption neighborhood j and worker type θ.

Individuals consume residential floor space, a tradable good, and a non-tradable good. We accommodate different preferences, productivities, wages, travel costs and consumption shares by type. The first term in the product in Equation (1) covers the utility factors specific to the worker's residence. She receives $b_n(\omega;\theta)$ in idiosyncratic residential amenities in neighborhood n, with mean level $B_n(\theta)$. Differences in residential preferences across groups will drive worker types to live in different neighborhoods. However, this worker dislikes more expensive residential floor space Q_n and tradable goods $P_n(T)$. We assume that the tradable good is traded without cost in the

[31] We abstract from tourists. Firstly, according to the Singapore Tourism Board, total tourist receipts made up 5.1% of total spending in the economy in 2015. If we remove accommodation receipts and expenditures on airfares/travel, this figure drops to 2.7%. This share is relatively small. Second, the travel patterns of tourists are unlikely to be affected by the changes in transit access from the opening of DTL. Airbnb is only available for long stays; furthermore, hotels and tourist attractions are concentrated downtown.

city and assume it is our numeraire: $P_n(T) = P(T) = 1$.

The second and third terms in the product in Equation (1) correspond to the utility terms specific to the worker's workplace and consumption neighborhood respectively. She earns a wage $w_i(G,\theta)$ per efficiency unit in neighborhood i. Meanwhile, she dislikes a more expensive price $P_j(N)$ of the non-tradable good in neighborhood j. To work in neighborhood i and consume non-tradables in neighborhood j, the worker pays iceberg travel costs $\exp(\kappa(\theta)\tau_{ni})$ and $\exp(\kappa(\theta)\tau_{nj})$ respectively. These travel costs increase with the time τ it takes to travel between neighborhoods. The parameters $\kappa(\theta)$ control the size of travel costs by worker group respectively.[32]

Next, high- and low-income workers have different consumption shares over housing, $\gamma(\theta)$, non-tradables, $\alpha(\theta)$, and tradables $1-\alpha(\theta)-\gamma(\theta)$ respectively. These shares measure the relative importance of housing, tradable and non-tradable consumption for utility for people of type θ.[33]

Finally, worker ω has idiosyncratic draws for residential amenities in neighborhood n, $b_n(\omega;\theta)$, efficiency units of labor (productivity) in neighborhood i and sector G, $a_i(\omega;G,\theta)$, and consumption amenities in neighborhood j, $s_j(\omega;\theta)$). We assume that these idiosyncratic shocks are drawn from the following independent Fréchet distributions:

$$\begin{aligned} F_n(b/B_n(\theta);R,\theta) &= \exp[-T_n(R,\theta)b^{-\varepsilon(R,\theta)}] \\ F_i(a;L,G,\theta) &= \exp[-T_i(\theta)T(G,\theta)a^{-\varepsilon(L,\theta)}] \\ F_j(s;S,\theta) &= \exp[-T_j(S,\theta)s^{-\varepsilon(S,\theta)}] \end{aligned} \tag{2}$$

where all the shape parameters, ε, are greater than 1; and all the scale parameters, T, are greater than zero. The distribution of idiosyncratic amenities is governed by the residential amenity level $B_n(\theta)$, which includes externalities from residential composition we will describe later.

The scale parameters, $T_n(R,\theta), T_i(\theta), T(G,\theta)$, and $T_j(S,\theta)$, control the overall level of the draws for residential preferences, work location productivity, sector productivity, and consumption preferences respectively. We allow scale parameters to vary across worker groups, to capture

[32] We expect differences in travel costs since low-income workers receive subsidies on travel. Higher travel costs imply larger gains from reductions in travel time.

[33] Larger consumption shares on housing imply larger implications of changes in rents for welfare. Larger consumption shares on non-tradables imply larger implications of access to consumption opportunities for welfare.

differences in preferences and productivities over locations across types. We also allow shape parameters, $\varepsilon(R,\theta), \varepsilon(L,\theta), \varepsilon(S,\theta)$, which control the dispersion of the distributions to differ across groups. A higher ε corresponds to a smaller dispersion of idiosyncratic taste shocks.[34]

Timing. Workers first choose where to live; then, in a second step, they choose where to work and where to consume non-tradables.[35] First, individuals observe their idiosyncratic residential amenities draws for all neighborhoods and choose to live in some neighborhood n. Second, individuals observe their idiosyncratic productivities for all workplaces and idiosyncratic consumption preferences in all neighborhoods, then choose to work in some neighborhood i and sector G, and to consume non-tradables in some neighborhood j. We solve the worker problem by backward induction.

4.1.1 Employment Location Decision

Having chosen a neighborhood n in which to live, individuals draw a vector of match-productivities with firms across the city as in Equation (2). With these draws in hand, workers choose to work in the neighborhood that offers the highest income net of their type-specific commuting costs:

$$\max_{i\in\mathbb{N},G\in\{N,T\}} \frac{w_i(G,\theta)a_i(\omega;G,\theta)}{\exp(\kappa(\theta)\tau_{ni})}. \tag{3}$$

Fix a worker type θ who has made the choice to live in neighborhood n. Since idiosyncratic productivities are distributed according to a Fréchet distribution, the probability she decides to work in neighborhood i and sector G is given by the following gravity equation:

$$\lambda_{niG}(L,\theta) = \frac{T_i(\theta)T(G,\theta)[w_i(G,\theta)\exp(-\kappa(\theta)\tau_{ni})]^{\varepsilon(L,\theta)}}{\sum_{l\in\mathbb{N}}\sum_{\tilde{G}=T,N} T_l(\theta)T(\tilde{G},\theta)[w_l(\tilde{G},\theta)\exp(-\kappa(\theta)\tau_{nl})]^{\varepsilon(L,\theta)}} \tag{4}$$

Individuals are more likely to travel to work in a neighborhood that pays a high wage net of travel costs, as in the numerator, relative to those in all other locations, as in the denominator.

[34] The sensitivity of choices to other variables such as travel costs is governed by the dispersion of preferences or productivity. When workers have similar matches in different locations (high ε), their choices are more sensitive to these other variables. Differences in heterogeneity across groups will be important in determining the incidence of travel costs, since these differences control the extent to which individuals are willing to bear high travel costs to work or consume in a location.

[35] Switching the order of workplace and consumption choice is without loss.

Differences in productivity heterogeneity across worker types is important in determining the incidence of travel costs, controlling the extent to which workers are willing to bear high commute costs to work in a neighborhood. Differences in productivity across neighborhoods and commute costs by type drive differences in work travel patterns across worker groups.

Expected income prior to drawing the vector of match productivities is directly related to the denominator in Equation (4) through

$$\mathbb{W}_n(\theta) = \Gamma\left(\frac{\varepsilon(L,\theta)-1}{\varepsilon(L,\theta)}\right)\left[\sum_{l\in\mathbb{N}}\sum_{\tilde{G}=T,N} T_l(\theta)T(\tilde{G},\theta)[w_l(\tilde{G},\theta)\exp(-\kappa(\theta)\tau_{nl})]^{\varepsilon(L,\theta)}\right]^{1/\varepsilon(L,\theta)} \tag{5}$$

where $\Gamma(.)$ is the Gamma function. Intuitively, in locations with better *access to employment* $\mathbb{W}_n(\theta)$, or access to locations with high expected income by type, workers are better off.

4.1.2 Consumption Location Decision

Having chosen to live in neighborhood n, workers draw a vector of idiosyncratic preference shocks with consumption locations across the city as in Equation (2). With these draws in hand, workers choose to consume non-tradables in the neighborhood with the best consumption amenities, accounting for travel costs and the price of non-tradables.

$$\max_{j\in\mathbb{N}} \frac{s_j(\omega;\theta)}{P_j(N)^{\alpha(\theta)}\exp(\kappa(\theta)\tau_{nj})} \tag{6}$$

Again, fix a worker type θ and residence location n. Since idiosyncratic consumption tastes $s_j(\omega;\theta)$ are distributed according to a Fréchet distribution, the probability this worker decides to consume non-tradables in j is given by the following gravity equation:

$$\lambda_{nj}(S,\theta) = \frac{T_j(S,\theta)[P_j(N)^{\alpha(\theta)}\exp(\kappa(\theta)\tau_{nj})]^{-\varepsilon(S,\theta)}}{\sum_{l\in\mathbb{N}} T_l(S,\theta)[P_l(N)^{\alpha(\theta)}\exp(\kappa(\theta)\tau_{nl})]^{-\varepsilon(S,\theta)}} \tag{7}$$

Individuals are more likely to travel to consume in neighborhoods with high consumption amenities net of the price of non-tradables and travel costs, as in the numerator, relative to those in all other locations, as in the denominator. Differences in preference heterogeneity across worker types is important in determining the incidence of travel costs, controlling the extent to which

workers are willing to bear high travel costs to consume non-tradables in a neighborhood. Differences in preferences across neighborhoods and travel costs by type drive differences in consumption travel patterns across worker groups.

Expected utility from non-tradable consumption prior to drawing the vector of idiosyncratic preferences is directly related to the denominator in Equation (7) through

$$\mathbb{S}_n(\theta) = \Gamma\left(\frac{\varepsilon(S,\theta)-1}{\varepsilon(S,\theta)}\right)\left[\sum_{l\in\mathbb{N}} T_l(S,\theta)[P_l(N)^{\alpha(\theta)}\exp(\kappa(\theta)\tau_{nl})]^{-\varepsilon(S,\theta)}\right]^{\alpha(\theta)/\varepsilon(S,\theta)} \tag{8}$$

where $\Gamma(.)$ is the Gamma function. Intuitively, in locations with better *access to consumption* $\mathbb{S}_n(\theta)$, or access to locations with low prices and high amenities by type, workers are better off.

4.1.3 Residential Location Decision

In the first stage, individuals choose where to live to maximize their expected indirect utility after observing their idiosyncratic residential amenity draws across all neighborhoods. Workers of type θ solve the following problem:

$$\max_{n\in\mathbb{N}} U_n(\omega;\theta) = \max_{n\in\mathbb{N}} \frac{b_n(\omega;\theta)}{Q_n^{\gamma(\theta)}}\mathbb{W}_n(\theta)\mathbb{S}_n(\theta) \tag{9}$$

Workers are attracted to locations with high residential amenities, low housing prices, high net incomes, and high utility from consumption of non-tradables.

Since idiosyncratic residence amenity draws are Fréchet distributed, the share of type-θ workers who live in neighborhood n is

$$\lambda_n(R,\theta) = \frac{T_n(R,\theta)[Q_n^{-\gamma(\theta)}B_n(\theta)\mathbb{W}_n(\theta)\mathbb{S}_n(\theta)]^{\varepsilon(R,\theta)}}{\sum_{l\in\mathbb{N}} T_l(R,\theta)[Q_l^{-\gamma(\theta)}B_l(\theta)\mathbb{W}_l(\theta)\mathbb{S}_l(\theta)]^{\varepsilon(R,\theta)}} \tag{10}$$

Similar residential amenities across neighborhoods (high ε) imply that choices are more sensitive to changes in access to employment, access to consumption, and rents. Differences in preferences across neighborhoods by type drive differences in residential patterns in the city across worker groups.

4.1.4 Expected Utility

Since we consider a "closed city", expected utility from living in the city prior to drawing the vector of idiosyncratic preferences is directly related to the denominator in Equation (10) through

$$\bar{U}(\theta) = \Gamma\left(\frac{\varepsilon(R,\theta)-1}{\varepsilon(R,\theta)}\right)\left[\sum_{l\in\mathbb{N}} T_l(R,\theta)[Q_l^{-\gamma(\theta)}B_l(\theta)\mathbb{W}_l(\theta)\mathbb{S}_l(\theta)]^{\varepsilon(R,\theta)}\right]^{1/\varepsilon(R,\theta)}. \tag{11}$$

4.2 Firms

4.2.1 Technology

In each location $i \in \mathbb{N}$, firms operate in both tradable and non-tradable sectors, $G \in \{N,T\}$. Sectoral productivity differs across locations. Firms produce under perfect competition and with a constant returns to scale technology. We specify firm production as being a Cobb-Douglas aggregate over labor and commercial floor space. Output is thus

$$Y_{iG} = A_{iG}L_{iG}^{\beta(G)}H_{iG}^{1-\beta(G)} \tag{12}$$

where $\beta(G) \in [0,1]$ and H_{iG} is commercial floor space. Labor input L_{iG} is a Cobb-Douglas aggregate over each worker group's effective units of labor, $\tilde{N}_{iG\theta}$:

$$L_{iG} = \tilde{N}_{iG}(+)^{\beta(G,+)}\tilde{N}_{iG}(-)^{1-\beta(G,+)} \tag{13}$$

where $\beta(G,+) \in [0,1]$ represents the intensity with which sector G uses high-type workers.[36] Each sector and neighborhood also has different productivities A_{iG}.

In location i, let P_{iG} be the price of variety G and W_{iG} the cost of labor in sector G.[37] Solving the firm's profit maximization problem, we find that the demand for labor and commercial floor

[36]The tradable sector requires a higher share of high-income workers, while the non-tradable sector heavily relies on low-income workers. Thus, the spatial distribution of non-tradable versus tradable production is strongly related to that of low-income versus high-income jobs. As a result, different worker types living in the same neighborhood may have different levels of access to employment opportunities.

[37]Derivations can be found in Online Appendix C.1.

space, as well as wages, can be written as

$$L_{iG} = \beta(G)\frac{P_{iG}Y_{iG}}{W_{iG}} \tag{14}$$

$$H_{iG} = [1-\beta(G)]\frac{P_{iG}Y_{iG}}{q_i} \tag{15}$$

$$\tilde{N}_{iG}(\theta) = \beta(G,\theta)\frac{L_{iG}W_{iG}}{w_i(G,\theta)} \tag{16}$$

4.3 Land Market

Following Combes, Duranton, and Gobillon (2019) and Epple, Gordon, and Sieg (2010), floor space is supplied by perfectly competitive developers; these developers are absentee landlords to whom land rents accrue.[38] Floor space is produced using land M_i and capital K_i, using a constant returns to scale technology:

$$\mathbb{H}_i = K_i^{\varphi} M_i^{1-\varphi} \tag{17}$$

where $\mathbb{H}_i$ is total floor space and φ is the share of land in floor space production. Cost minimization implies that

$$Q_i = \mathbb{P} M_i^{\frac{\varphi-1}{\varphi}} \mathbb{H}_i^{\frac{1-\varphi}{\varphi}} \varphi^{-1}, \tag{18}$$

where $\mathbb{P}$ is the common price for capital across all neighborhoods.[39] As the production of non-tradable goods and the demand for floor space rise in response to increased consumption travel, increased rents may drive out tradable production and residents, shifting commercial and residential spatial patterns across the city.[40]

Following Ahlfeldt et al. (2015), after factoring in the tax equivalent of land use regulations, the land market equilibrium requires no arbitrage between the commercial and residential use of

[38] Singapore's housing subsidies are broad-based and over 80% of the population is eligible. These subsidies are standardized across the country.

[39] Derivations can be found in Online Appendix C.3. We do not need to solve for the common price for capital $\mathbb{P}$ in the initial equilibrium because our counterfactual analysis uses exact-hat algebra.

[40] In robustness checks, we consider the impact of the DTL when housing is perfectly price inelastic; we find similar qualitative results to our main analysis. See Section 6.2.1 for details.

floor space. The commercial price of floor space for both the tradable and non-tradable sector is

$$q_i = \xi_i Q_i, \tag{19}$$

where ξ_i equals one plus the tax equivalent of land use regulations that restrict commercial land use relative to residential land use. We allow this wedge between commercial and residential floor prices to vary across neighborhoods.

4.4 Market Clearing

Land. Summing over housing expenditures for all residents, the expression for demand for residential floor space in neighborhood n is

$$H_n(R) = Q_n^{-1} \sum_{\theta} \gamma(\theta) R(\theta) \lambda_n(R, \theta) \mathbb{W}_n(\theta) \tag{20}$$

Market clearing for floor space requires that the total supply of floor space equals the total floor space demanded from both residents and firms in each neighborhood j:

$$\mathbb{H}_j = H_j(R) + \sum_{G \in \{N,T\}} H_{jG}. \tag{21}$$

Labor. Using the commuting probabilities from Equation (4), the supply of workers to any location is the sum over the number of commuters by origin that travel to that work location:

$$N_{iG}(\theta) = \sum_n \lambda_{niG}(L, \theta) \lambda_n(R) R(\theta) \tag{22}$$

Labor supply in the model takes a log-linear form that depends on two forces. Firstly, more workers commute to destinations paying higher wages. Second, firms attract workers when they have better access to them through the commuting network. Ultimately, workers care about wages net of commute costs. Meanwhile, total effective labor supply to location i is given by

$$\tilde{N}_{iG}(\theta) = \sum_n R_n \lambda_{niG}(L, \theta) \bar{a}_{ni}(G, \theta) \tag{23}$$

where $\bar{a}_{ni}(G, \theta)$ is the average productivity of type-θ workers who live in n and decide to work in

i.[41]

In each neighborhood i and sector G, market clearing requires that the supply of effective labor, as in Equation (23), equals the demand for effective labor, as in Equation (14). Wages by worker type are endogenously determined by market clearing.

Non-tradables. In each neighborhood j, total receipts for non-tradables must equal total expenditures on non-tradables:

$$P_j(N)A_{jN}L_{jN}^{\beta(N)}H_{jN}^{1-\beta(N)} = \sum_{n\in\mathbb{N}}\sum_{\theta\in\{+,-\}} \alpha(\theta)\lambda_{nj}(S,\theta)\lambda_n(R,\theta)R(\theta)\mathbb{W}_n(\theta). \tag{24}$$

Prices of non-tradables are endogenously determined to clear the market. We sum over expenditures of non-tradables for all the workers who travel from some neighborhood n to consume in neighborhood j across worker groups.

4.5 Externalities

4.5.1 Productivities

A location's productivity depends on both an exogenous component $\bar{A}_{iG}$ that reflects features independent of economic activity (e.g. access to roads, slope of land) as well as the endogenous density of employment in that location:

$$A_{iG} = \bar{A}_{iG}\left[\bar{H}_i^{-1}(N_i(+)+N_i(-))\right]^{\mu(A)}, \tag{25}$$

where $\bar{H}_i$ is the total units of land in location i. The strength of agglomeration externalities is governed by the parameter $\mu(A)$.[42]

[41] Average productivity $\bar{a}_{ni}(G,\theta)$ follows from the Fréchet distribution of idiosyncratic worker productivities. See Online Appendix Section C.2.

[42] We permit spillovers in productivity by sector but disallow them from being type-specific. We collapse productivity spillovers by type for parsimony. Absent spillovers of any kind, our model generates qualitatively similar counterfactual predictions. See Table 3. Type-specific productivity spillovers would lead workers of the same (income, sector) to concentrate in neighborhoods in which they are already abundant.

4.5.2 Amenities

Amenities in a neighborhood are a function of an exogenous component $\bar{B}_n(\theta)$ and a residential externality. This externality depends on the ratio of high-skilled to low-skilled residents:

$$B_n(\theta) = \bar{B}_n(\theta) \left(\frac{R_n(\theta)}{R_n(\neg\theta)} \right)^{\mu(U,\theta)} \tag{26}$$

In our model, instead of being a function of the total density of residents, endogenous amenities depend on the demographic composition of workers across income groups, similar to Tsivanidis (2019).[43] Workers are more willing to pay to live in neighborhoods that are high in amenities of their type. As workers locate in these neighborhoods, they increase these type-specific endogenous amenities even more, strengthening segregation.

4.6 Equilibrium

We now define the general equilibrium of our model.

Definition Given vectors of exogenous location characteristics $\{\bar{B}_n(\theta), \bar{A}_{iG}, \tau_{ij}, \bar{H}_j, \psi_i, \xi_i\}$, city group-wise populations $R(\theta)$, and model parameters

$$\{\alpha, \beta, \gamma, \kappa(\theta), T_n(R,\theta), T_i(\theta), T(G,\theta), T_j(S,\theta), \varepsilon(R,\theta), \varepsilon(L,\theta), \varepsilon(S,\theta), \varphi, \mu(A), \mu(U,\theta)\},$$

an *equilibrium* is a vector of endogenous objects

$$\{q_i, w_i(G,\theta), H_{iG}, \tilde{N}_{iG\theta}, \lambda_n(R,\theta), \lambda_{niG}(L,\theta), \lambda_{nj}(S,\theta), \bar{U}(\theta)\}$$

such that:

1. **Labor Market Clearing**: The supply of labor by individuals in Equation (23) is consistent with demand for labor by firms in Equation (14).

[43] In contrast to the literature on neighborhood spillovers in the United States (e.g., Guerrieri, Hartley, and Hurst 2013), in Singapore, there is qualitative evidence that low-income workers prefer to live in neighborhoods with others of similar socioeconomic status. A stark example comes from an ethnographic study of inequality (Y. Y. Teo 2022, pp. 57, 63). Low-income apartments in Singapore "are not ghettoized spaces... [and] there are no slums. In terms of access to clean water, electricity, amenities, and transportation, the people [the author] meet[s] are not denied access to these things in any absolute or physical way." Yet "a single mother once told [the author] she is afraid to walk through a mall near her home... because her six-year-old daughter will ask her to 'buy this, buy that.' "

2. **Floor Space Market Clearing**: The market for floor space clears as in Equation (21), its price is consistent with residential populations (20) and commercial floor space use (15), and total floor space is consistent with land developer optimality (18).

3. **Goods Market Clearing**: Non-tradable consumption matches non-tradable production in each neighborhood, as in Equation (24).

4. **Closed City**: Populations add up to the city total.

We discuss equilibrium existence and uniqueness. In a version of the model without externalities, by similar arguments to those made in Ahlfeldt et al. (2015), the model's congestion forces — commuting costs, travel costs to consume non-traded goods, and an inelastic land supply — ensure that a unique equilibrium exists. In the full version of the model, by adapting arguments in Section 3.1 of Allen, Arkolakis, and Li (2022), uniqueness can be guaranteed when "agglomeration forces are small relative to congestion forces."[44]

4.7 Model Discussion

First of all, our model captures the direct distributional implications of changing access to consumption, in addition to changing employment access. Transit improvements affect how easily different worker types access consumption opportunities, and these effects also depend on where they (want to) live, work and consume. Suppose travel costs fell between low-income neighborhoods and locations in which non-tradable goods are cheap and high in (type-specific) quality. Then low-income workers would become better off. Access to consumption is more important to welfare for worker types with 1) larger expenditures on non-tradables, 2) larger dispersion of preferences over consumption locations (more inelastic demand), and 3) higher travel costs.

[44] Characterizing formal sufficient conditions for uniqueness involves tedious algebra. We sketch the approach here. First, bound the absolute values of the elasticities of productivity spillovers with respect to worker population; and those of amenity spillovers with respect to residential population. Next, substitute the gravity and spillover equations into the market clearing equations in residential population, labor supply, and consumption travel. This substitution gives rise to bounds on the elasticities of populations in each workplace, residence, and consumption location by type. Collect these elasticity bounds in a matrix of "cross-elasticities." Then, by Remark 1 and part (i) of Theorem 1 of Allen, Arkolakis, and Li (2022), for equilibrium uniqueness, it is sufficient that the spectral density of the matrix of "cross-elasticities" be less than one.

Next, consumption travel shapes the residential patterns of the city in a type-specific manner. Workers trade off access to non-tradables, along with access to employment and higher rents when choosing where to live. Worker types with less dispersion in residential amenities are better able to take advantage of both improved access to consumption and employment in other locations. In contrast with that of Miyauchi, Nakajima, and S. Redding (2022), our specification of consumption travel preserves tractability while still allowing us to speak to differences in behavior between low- and high-income workers.

Lastly, incorporating travel to consume non-tradables also has distributional implications for access to employment across worker groups. Responding to changes in transit access, consumers change where they consume non-tradables, trading off travel costs and type-specific amenities and prices. Thus relative factor demand changes across locations. Since firms selling non-tradables hire more low-income workers, low-income jobs move to where residents are traveling more for consumption (thus demanding more non-tradable production in those locations). High-income jobs in tradables may also move away from locations with greater consumption travel, since commercial rents will rise in response to greater production of non-tradables. Therefore, low-income jobs could move to locations closer or further from where low-income workers live, locations with higher or lower wages for low-income workers, or locations which are more or less productive for low-income workers, and analogously for high-income workers. Since commercial activity between tradable and non-tradable sectors reorganizes across space, each worker group faces differential changes in expected access to employment or expected income net of commuting. Therefore, it is an empirical question whether inequality rises or falls.[45]

5 Estimation

This section estimates the model from Section 4.

First, using gravity specifications from the model, we estimate workplace and consumption preferences and access across neighborhoods and across worker groups. In this first step, we exploit two sources of variation. One source is cross-sectional: we see variation in travel times and

[45] These reallocative predictions of the model could be directly tested with data on exogenous shocks to demand for non-tradable goods.

variation in commuting (or consumption travel) flows between residence-destination pairs by income type. The other source of variation is generated by the opening of stage 2 of the Downtown Line (DTL2): differences between flows shortly before and after the line was opened identify the impact of lower travel times on commuting (or consumption travel) choices. Meanwhile, differences between flows in the long run before and after DTL2 was opened identify changes in the spatial (re-)organization of economic activity.

Second, we estimate type-specific work and consumption travel costs, as well as parameters governing the dispersion of idiosyncratic workplace productivities and consumption amenities. For workplace productivity, by combining variation in residential incomes by type across neighborhood with variation in the estimated residence fixed effects in the first step, we can identify the dispersion of idiosyncratic workplace productivities by type, $\hat{\varepsilon}(L,\theta)$. Using the estimated values of $\hat{\varepsilon}(L,\theta)$ and the semi-elasticities for consumption and work travel, we can recover the dispersion of idiosyncratic consumption amenities by worker type and the travel costs per worker type.

Third, we estimate parameters that govern the dispersion of idiosyncratic residential amenities and residential spillovers by type. When DTL2 opened, travel times fell. In the long run, workers and firms re-optimize, shifting the spatial distribution of economic activity in Singapore. Therefore, variation in residence shares, rents, market access, and residential composition across time identifies amenity spillovers across space.

Fourth, we recover residential amenities across neighborhoods and across high- and low-income workers. We validate that the model-implied residential amenities are highly correlated with external data, including the spatial distribution of schools, community clubs, and parks.

Fifth, we calibrate the remaining parameters of the model.[46]

5.1 Estimating Consumption Preferences and Access

First, we estimate workplace and consumption preferences and access using data from before the DTL opened. For each worker type $\theta \in \{+,-\}$, we combine the consumption travel flow Equation

[46] Our model can be inverted to obtain unobservable endogenous variables that rationalize the observed data as an equilibrium of the model. See Online Appendix D.

(7) with the specification of travel costs $\phi(S,\theta) = \varepsilon(S,\theta)\kappa(\theta)$. The resulting gravity equation relates the variation in consumption travel flows to the variation in travel times:

$$\ln \lambda_{nj}(S,\theta) = \eta_j(S,\theta) - \mu_n(S,\theta) - \phi(S,\theta)\tau_{nj} + u_{nj}(S,\theta). \tag{27}$$

In this equation, $\eta_j(S,\theta)$ and $\mu_n(S,\theta)$ are workplace and origin fixed effects, and $u_{ni}(S,\theta)$ is an unobserved shifter of consumption travel propensities. We take τ_{nj} as the average time taken to travel between neighborhoods n and j in the farecard data, and construct work travel shares from identified residences and consumption trips in the farecard data.

We note that travel times τ may be correlated with the unobservable $u_{nj}(S,\theta)$. Endogeneity in this setting takes two forms. The first is conventional: transit development may be driven precisely by where workers travel to consume non-tradables. To deal with this source of endogeneity, in our preferred specification, we instrument for τ with the straight-line distance between neighborhood centroids. Columns 3 and 4 of Table 1 report the results from the regression associated with Equation (27). We estimate a negative and statistically significant semi-elasticity of consumption travel flows with respect to travel time. These semi-elasticities take the value of $\hat{\phi}(S,+) = -0.053$ for high-income workers and $\hat{\phi}(S,-) = -0.094$ for low-income workers. Thus, we find that low-income workers are more sensitive to changes in travel time $(p < 0.01)$. In the next subsection, we will show that, across groups, their respective gravity coefficients are larger for consumption travel than work travel.

The second source of endogeneity is from possible sorting and complementarities across (origin, consumption destination) pairs (see, e.g., Dingel and Tintelnot 2020 and Severen 2023). To assess robustness of our preferred specification to such complementarities, we also estimate Equation (27) using a difference-in-differences framework:

$$\Delta \ln \lambda_{nj}(S,\theta) = \Delta\eta_j(S,\theta) - \Delta\mu_n(S,\theta) - \phi(S,\theta)\Delta\tau_{nj} + \Delta u_{nj}(S,\theta), \tag{28}$$

where the differences are across trips taken before and after the line opening. Identification of $\phi(S,\theta)$ then requires $\mathbb{E}[\Delta\tau_{nj} \times \Delta u_{nj}(S,\theta)] = 0$, i.e., that sorting patterns and flow correlations

across distance are stable.

Instead of pooling across time, this difference-in-differences method enables us to leverage two "intermediate" control groups of origin-destination *pairs*, where one of the two "ends" is treated and the other is not. Consider workers living in Bishan, an origin neighborhood not sited on top of a DTL2 station. Workers consuming non-tradable goods in Beauty World (who benefit from DTL2) are compared with those who visit Orchard (who do not).[47] Similarly, consider Orchard, a major shopping belt not served directly by DTL2. Shoppers who live in Bishan (not on DTL2) are compared with those who live in Bukit Panjang (a terminus of DTL2). These comparisons "control for many potential unobserved motives for changing travel behavior" (Severen 2023).

The exclusion restriction would be violated if, for instance, after DTL2 opened, shops in all neighborhoods served by DTL2 systematically favored demographic groups prevalent in the far west of Singapore (hence closer to DTL2 neighborhoods) relative to those in the east. Such a violation seems implausible because Singapore is highly racially and socioeconomically integrated.

In columns 3 and 4 of Table A6, we display the results from the regression associated with Equation (28). We find that the semi-elasticities recovered in this alternate procedure are quantitatively similar to those recovered when estimating the first equation (in levels).

The consumption location fixed effect $\eta_j(S,\theta)$ from the first regression gives us measures of consumption attractiveness across neighborhoods:

$$\exp(\hat{\eta}_j(S,\theta)) = T_j(S,\theta)P_{jN}^{-\alpha(\theta)\times\varepsilon(S,\theta)}, \tag{29}$$

which captures average consumption amenities by type, as well as the prices of non-tradables in each neighborhood i. In Figure A17a, we display a scatter plot of consumption preferences by high- and low-income workers. We observe substantial differences in where low and high income groups prefer to consume non-tradables. As a test of model fit, in Table A8, for each income group, we display regression coefficients of consumption attractiveness against various external measures

[47] The fastest transit route from Bishan to Beauty World is to take the Circle Line from Bishan to the Botanic Gardens, then transfer to the Downtown Line headed northwest to Beauty World. In contrast, Orchard is accessible from Bishan via the North-South Line, sans transfers.

of retail amenities. We find that our attractiveness measure is positively correlated with the density of food establishments, supermarkets and clinics in a neighborhood.[48]

The residence fixed effect term $\mu_n(S,\theta)$ from the above regression gives us

$$\exp(\hat{\mu}_n(S,\theta)) = \sum_{m\in\mathbb{N}} T(S,\theta)_m P_m(N)^{-\alpha(\theta)\times\varepsilon(S,\theta)} \exp(-\varepsilon(S,\theta)\times\kappa(\theta)\times\tau_{nm}) \tag{30}$$

which, for fixed $\varepsilon(S,\theta)$ and origin neighborhood n, allows us to recover our measure of consumption access, $\mathbb{S}_n(\theta)$, across neighborhoods. This measure reflects the expected utility from non-tradable consumption across destination neighborhoods j.

5.2 Estimating Workplace Preferences and Access

Next, we estimate workplace preferences and access. For each worker type $\theta \in \{+,-\}$, we combine the work travel flow Equation (4) with travel costs $\phi(L,\theta) = \varepsilon(L,\theta)\kappa(\theta)$. The resulting gravity equation relates the variation in work travel flows to the variation in travel times:

$$\ln\lambda_{ni}(L,\theta) = \eta_i(L,\theta) - \mu_n(L,\theta) - \phi(L,\theta)\tau_{ni} + u_{ni}(L,\theta) \tag{31}$$

where $\eta_i(L,\theta)$ and $\mu_n(L,\theta)$ are fixed effects, and $u_{ni}(L,\theta)$ is an unobserved shifter of work travel propensities. As before, we let τ be the average travel time between neighborhoods in the farecard data, and construct work travel shares from identified residences and workplaces in the farecard data.

As for the consumption gravity equation, travel times τ_{ni} may be correlated with the unobservable $u_{ni}(L,\theta)$. Again, we take two approaches to deal with endogeneity. First, travel times may be correlated with unobserved variation in transit development that affects workplace choice. To deal with this issue, we instrument for τ with the straight-line distance between neighborhood centroids. Columns 1 and 2 of Table 1 report the results from estimating Equation (31). We estimate a negative and statistically significant semi-elasticity of work travel flows with respect to travel time

[48] Interestingly, we find that low-income workers prefer locations where low-cost options make up a larger share of food establishments, such as hawker stalls and food courts, while the corresponding coefficient is insignificant at the 5% level for high-income workers. This last finding corroborates our descriptive evidence that the rich and poor consume different non-tradable goods.

of $\hat{\phi}(L,+) = -0.051$ for high-income workers and $\hat{\phi}(L,-) = -0.070$ for low-income workers. Again, we find a gravity coefficient for low-income workers that is larger in magnitude ($p < 0.01$), indicating that low-income workers are more sensitive to changes in travel time. Comparing our estimates for work trips to those of consumption trips, we find smaller gravity coefficients for work travel than consumption travel across both groups.

As before, a second source of endogeneity is from possible sorting and complementarities across (origin, workplace destination) pairs. To assess robustness of our preferred specification to such complementarities, we also estimate Equation (32) using a difference-in-differences framework:

$$\Delta \ln \lambda_{ni}(L,\theta) = \Delta\eta_i(L,\theta) - \Delta\mu_n(L,\theta) - \phi(L,\theta)\Delta\tau_{ni} + \Delta u_{ni}(L,\theta), \tag{32}$$

where the differences are taken across trips taken before and after the line opening. Identification of $\phi(L,\theta)$ then requires $\mathbb{E}[\Delta\tau_{ni} \times \Delta u_{ni}(L,\theta)] = 0$, i.e., that sorting patterns and flow correlations across distance are stable. Similar arguments as in the previous subsection apply.[49]

In columns 1 and 2 of Table A6, we display the results from the regression associated with Equation (32). Again, we find that the semi-elasticities recovered in this alternate procedure are quantitatively similar to those recovered when estimating the first equation (in levels).

The workplace fixed effect from estimation gives us measures of workplace attractiveness across neighborhoods:

$$\exp(\hat{\eta}_i(L,\theta)) = \sum_{G\in\{N,T\}} T_i(\theta)T(G,\theta)w_{iG}(\theta)^{\varepsilon(L,\theta)}, \tag{33}$$

which captures type-specific averages, across sectors, of productivity and wages in each neighborhood i. Figure A17b presents a scatter plot of workplace preferences by high- and low-income workers. We observe substantial differences in where low and high income groups prefer to work.

[49]The exclusion restriction is violated if, for instance, after DTL2 opened, workplaces in all neighborhoods served by DTL2 systematically favored demographic groups prevalent in the far west of Singapore relative to those in the east. Such a violation seems implausible because of laws against discrimination in hiring, in addition to the fact that Singapore is highly racially and socioeconomically integrated.

In Table A7, we find that workplace attractiveness for both groups is highly correlated with employment density.

The residence fixed effect from estimation gives us

$$\exp(\hat{\mu}_n(L,\theta)) = \sum_{l\in\mathbb{N}} \sum_{G\in\{N,T\}} T_i(\theta)T(G,\theta)w_{iG}(\theta)^{\varepsilon(L,\theta)} \exp(-\varepsilon(L,\theta)\times\kappa(\theta)\times\tau_{nl}). \tag{34}$$

Given idiosyncratic shocks $\varepsilon(L,\theta)$ and an origin neighborhood n, we can recover our measure of employment access, $\mathbb{W}_n(\theta)$. This measure captures the expected wages net of commuting across neighborhoods i and sectors G.

5.3 Parameter Estimation: Travel Elasticities and Costs

5.3.1 Dispersion of Workplace Idiosyncratic Productivity

Taking logs of the expression for expected income by worker type in Equation (5), we can estimate the workplace dispersion parameter, $\varepsilon(L,\theta)$, by regressing the estimated residence fixed effects, $\hat{\mu}_n(L,\theta)$, on the log of average residential income from observed administrative data by type. The parameter $\varepsilon(L,\theta)$ is thus identified off variation across neighborhoods in income and the estimated measure of residence attractiveness.[50]

$$\hat{\mu}_n(L,\theta) = \varepsilon(L,\theta)\ln(\text{Residential Income}_n) + e_n \tag{35}$$

The residual e_n accounts for estimation error in the residence fixed effects and any departures from model assumptions.

Table 2 presents our estimation results. We find a strong positive correlation between our model's estimate of expected income with that of non-targeted data. We show in Figure A18, similar to what Miyauchi, Nakajima, and S. Redding (2022) find, that residence fixed effects are generally log-linear in residential income. However, there are outliers in a few high-income neighborhoods, reflecting non-labor income at high income levels. In our preferred specification, we drop

[50]Since we have rich transit data, had we a panel of wages across neighborhoods, we would estimate a differenced version of Equation (35). Then, identification only requires stability of sorting patterns across neighborhoods. Unfortunately our wage data are from 2010 and 2015, with the former being before our sample period.

the four richest neighborhoods, estimating a workplace idiosyncratic dispersion of $\hat{\varepsilon}(L,+) = 2.912$ for high-income workers and $\hat{\varepsilon}(L,-) = 5.023$ for low-income workers respectively. Consistent with findings in Tsivanidis (2019), our point estimates suggest that high-income workers may have more inelastic work travel patterns than low-income workers.[51] Therefore, high-income workers may benefit more from reductions in travel costs, being less able to substitute to more attractive work locations in equilibrium.

5.3.2 Dispersion of Idiosyncratic Consumption Amenities

Next, we can compute the consumption travel dispersion parameter by worker type, $\varepsilon(S,\theta)$, using the estimated semi-elasticities for consumption and work travel, $\phi(S,\theta)$ and $\phi(L,\theta)$:

$$\hat{\varepsilon}(S,\theta) = \hat{\varepsilon}(L,\theta)\hat{\phi}(S,\theta)/\hat{\phi}(L,\theta) \tag{36}$$

We find a idiosyncratic consumption dispersion of $\hat{\varepsilon}(S,-) = 3.00$ for high-income workers and $\hat{\varepsilon}(S,-) = 5.81$ for low-income workers. See the first panel of Table A9 for the estimated parameters. As in the previous section, our point estimates suggest that high-income workers have more inelastic consumption travel patterns than low-income workers. Thus, high-income workers could benefit more from reductions in travel costs as they are less able to substitute to more attractive consumption locations in equilibrium. Idiosyncratic dispersion in consumption appears slightly larger for consumption trips than workplace trips.

5.3.3 Travel Costs

Finally, we can back out the travel cost parameter by worker type:

$$\hat{\kappa}(\theta) = \hat{\phi}(S,\theta)/\hat{\varepsilon}(S,\theta) \tag{37}$$

[51] In robustness exercises considered in Section 6.3, we consider how our inequality predictions change if both income groups had (1) the same travel costs; or (2) the same travel elasticities; or (3) the same residential elasticities. We find similar qualitative results under these alternate assumptions.

More data are needed to confirm a difference in workplace idiosyncratic dispersion between high- and low-income workers. A path forward is to link income *by worker* to their choice of residence and workplace, which would be possible with data from, e.g., the Central Provident Fund of Singapore.

We estimate a travel cost parameter of $\hat{\kappa}(+) = 0.018$ for high-income workers and $\hat{\kappa}(-) = 0.014$ for low-income workers, as displayed in the first panel of Table A9. Our point estimates suggest that low-income workers have slightly lower travel costs, with the wedge between travel costs consistent with the 25% subsidy low-income workers receive on transit fares.[52]

5.4 Parameter Estimation: Dispersion of Idiosyncratic Residential Amenities and Externalities

To estimate spillovers in amenities and production, we exploit the fact that changes in market access induced by the Downtown line result in a shock to the supply of labor and resident locations across the city. To operationalize this fact, we estimate the model in 2015 and 2018 before and after the opening of the DTL.[53] By log-linearizing Equation (10) and substituting in Equation (26), we can estimate $\mu(U,\theta)$ and $\varepsilon(R,\theta)$ with the following equation:

$$\Delta \ln \lambda_n(R,\theta) = \varepsilon(R,\theta)\Delta \ln Q_n^{-\gamma(\theta)}\mathbb{W}_n(\theta)\mathbb{S}_n(\theta) + \varepsilon(R,\theta)\mu(U,\theta)\Delta \ln \left[\frac{R_n(\theta)}{R_n(\neg\theta)}\right] + \Delta \ln e_n(\theta), \tag{38}$$

where Δ indicates the difference in estimated model unobservables (or data) before and after the line opening. Identification of $\varepsilon(R,\theta)$ and $\mu(U,\theta)$ requires that changes in local rents and access are orthogonal to changes in latent residential amenities, controlling for residential composition.[54]

We present the estimates from Equation (38) in the second panel of Table A9. Our point estimates suggest that the dispersion of idiosyncratic residential amenities is slightly larger for high types at $\varepsilon(R,+) = 1.48$ than that of low types at $\varepsilon(R,-) = 1.38$. High-income workers may be better able to take advantage of improvements in access across neighborhoods by shifting

[52] The ratio of travel costs between low- and high-income workers is $0.014/0.018 = 0.778$.

[53] We use housing prices from 2008, just before the precise station locations on the DTL were announced. In our descriptive analysis in Subsection 3.2, we showed that price appreciation, between control apartments and treated apartments near the planned line, diverged after 2008, but not before. Were we to use prices from 2015, much of the improved accessibility from the line would have already been "priced in."

[54] While our identification assumption is strong, our main results do not hinge on the dispersion of spillovers in amenities and production. In Table 3, we report our counterfactual estimates in a model without spillovers, finding no qualitative difference compared to our preferred specification.

where they live.[55] We find stronger externalities for low types ($\mu(U,-) = 0.45$) than high types ($\mu(U,+) = 0.22$), consistent with the findings from Tsivanidis (2019). This finding implies that there are stronger spillovers in endogenous amenities for low-income workers, and hence a stronger force towards clustering in the same neighborhoods.

5.5 Estimating Residential Amenities

We define composite residential amenities $\mathbb{B}_n(\theta)$ and recover them using known parameters, observed land prices and travel shares, and estimated measures of work and consumption access.

$$\mathbb{B}_n(\theta) = B_n(\theta) \times (T_n(R,\theta))^{1/\varepsilon(R,\theta)} = \frac{Q_n^{\gamma}\lambda_n(R,\theta)^{1/\varepsilon(R,\theta)}\bar{U}(\theta)}{\mathbb{W}_n(\theta)\mathbb{S}_n(\theta)} \times \Gamma\left(\frac{\varepsilon(R,\theta)-1}{\varepsilon(R,\theta)}\right)^{-1} \tag{39}$$

In Table A10, we show that composite residential amenities are highly correlated with external data on amenities — specifically schools, community clubs, and parks. Both high- and low-income residents care about community clubs, schools, and parks.

5.6 Calibrated Parameters

Calibrated values are reported in Table A11 in the Online Appendix. The consumption shares, (α,γ), for low- and high-income groups (below and above the 20th percentile) are calibrated to match line-itemized expenditures data from the 2018 Singapore Household Expenditure Survey.[56] These expenditures data are available by income bracket. The overall labor share by sector, $\beta(G)$, is matched to estimates from the 2013 Economic Survey of Singapore. Labor shares by type, $\beta(G,\theta)$, are calibrated to the average share of the wage bill paid to low- and high-income workers in Singapore by sector using the 2018 Singapore Labor Force Survey. We assume a share of land in construction costs of $\varphi = 0.25$ following Epple, Gordon, and Sieg (2010), Combes, Duranton,

[55] Similar to the dispersion in workplace productivity, better income data is needed to confirm this difference.

[56] See Table A3 for a detailed breakdown of expenditures on non-tradeable goods and services by worker group. We only include expenditures on goods and services which are tied closely to urban public transit trips. For goods, we restrict to expenditures on retail products and use data from a 2017 PricewaterhouseCoopers survey on consumer preferences for in-store shopping by product to classify goods where over 50% of respondents prefer to purchase it in-person as non-tradable.

and Gobillon (2019) and Ahlfeldt et al. (2015). We consider production agglomeration parameters consistent with Rosenthal and Strange (2004), Melo et al. (2009), and Ahlfeldt et al. (2015).

6 Counterfactuals

In this section, we describe the counterfactual scenarios we study. In our main analysis, we compute the impact of the Downtown Line on inequality and welfare.[57]

This section proceeds as follows. First, we outline our exact-hat approach. Second, we test our counterfactual predictions on ex-post data. Finally, we present our main results.

6.1 Exact-Hat Algebra

We undertake counterfactuals with an exact-hat approach (Dekle, Eaton, and Kortum 2008). Rather than estimating our model in terms of levels, we specify the model in terms of changes from the current equilibrium, thereby finessing having to assemble proxies for various unobservables in our model.[58] We let $\hat{x} = x'/x$ denote the relative change in the equilibrium endogenous variable x, where x' is the value of x in the new equilibrium. We consider an exogenous change in travel times $\Delta\tau_{ij} = \tau'_{ij} - \tau_{ij}$. Given model parameters $\{\alpha, \beta, \gamma, \kappa(\theta), \varepsilon(R,\theta), \varepsilon(L,\theta), \varepsilon(S,\theta), \varphi, \mu(A), \mu(U,\theta)\}$, assumed bilateral changes in travel times, $\{\exp(\Delta\tau_{ij})\}$, and endogenous variables in the initial equilibrium $\{\lambda_{nj}(S,\theta), \lambda_{ni}(L,\theta), \lambda_n(R,\theta), H_{nG}, H_n(R), \mathbb{W}\}$, we solve the system of "hat equations"[59] starting with an initial guess in each endogenous variable such that $\hat{x} = 1$, updating our guess till the algorithm has converged to an equilibrium. With the counterfactual changes in endogenous variables $\{\hat{B}_n(\theta), \hat{\mathbb{S}}_n(\theta), \hat{\mathbb{W}}_n(\theta), \hat{Q}_n\}$, the change in expected utility by type is:

$$\hat{\hat{U}}(\theta) = \left(\sum_{n \in \mathbb{N}} \lambda_n(R,\theta) \left(\hat{B}_n(\theta) \hat{\mathbb{S}}_n(\theta) \hat{\mathbb{W}}_n(\theta) \hat{Q}_n^{-\gamma(\theta)} \right)^{\varepsilon(R,\theta)} \right)^{1/\varepsilon(R,\theta)} \tag{40}$$

We estimate the initial equilibrium using data from 2015, then use changes in travel times as

[57] For data availability reasons, we focus on Phase 2 and 3 of the line. These two phases comprise 80% of the line and connect the outer neighborhoods of the city to downtown (as opposed to Phase 1, a short "test" segment downtown).

[58] These unobservables include the prices of non-tradable goods, neighborhood-sector wages per efficiency unit by type, mean idiosyncratic draws, and so on.

[59] Equations are available in Online Appendix E.

observed in the fare card data from before the opening of the DTL in 2015 to after its opening in 2018. Figure A6 presents the distribution of the changes in travel time.[60]

6.2 Welfare and Inequality

6.2.1 Main Results

In the top panel of Table 3, column 1 presents the effect of the Downtown line on welfare. This effect is broken down by access to employment and consumption by high- and low-income types respectively. We find that the DTL improves welfare, $\bar{U}(\theta)$, for high income workers, but not for low-income workers. Welfare increases by 1.8% for high-income workers, while welfare for low-income workers remains stagnant at a 0.1% increase. Although consumption access increases for both groups, non-tradable production and low-income jobs move to less attractive locations. Access to consumption, $\mathbb{S}$, improves for both workers, with high income workers benefiting by 1.4%, higher than low-income workers at 1.0%. However, while access to employment (expected wages net of commuting costs), $\mathbb{W}$, increases for high-income workers by 1.11%, low-income workers experience a 1.13% decline. Overall, high income workers experience a 1.7 p.p. larger increase in welfare compared to low income workers.[61]

There are two main mechanisms driving our results. First, the DTL disproportionately improves access to residents of many high-income areas. Figure 1 shows that the Downtown Line directly serves the many high income areas in a cluster just west of the city center. This partially explains the larger improvements in both access to consumption and employment experienced by high-income workers relative to low-income workers. Second, in response to the DTL, a greater

[60] In Online Appendix F.1, we test our model's predictions on data from 2018. We find that predicted changes in travel patterns and residential shares are highly correlated with observed post-DTL changes. Furthermore, predicted changes in land use for non-tradable good production are correlated with post-DTL food establishment entry between 2015 to 2018. Thus, the model has predictive power over changes in travel patterns, residence choices and commercial activity in response to the DTL.

[61] Our results are qualitatively similar if housing supply were assumed to be perfectly price inelastic; see Table A13. The main results are also robust to the absence of spillovers (see the bottom panel of Table 3), as well as the "covariates-based approach" of Dingel and Tintelnot (2020). The results are also similar for an extension of the model with roundabout production where tradable goods are used as an input into production, following Caliendo and Parro (2015) (see Table A15). Additionally, we present results assuming the DTL only reduces consumption travel time while holding commute times constant (see Table A15). See Online Appendix F for details.

number of consumers travel to consume non-tradables near the center of the city. These consumers benefit from improved access to downtown via the new line. Figure 4 plots the model predictions for changes in consumption trips across neighborhoods. We see a large increase in consumption trips made to the center of the city. As a result, non-tradable jobs, disproportionately low-income, move to the center of the city; meanwhile, low-income workers live far away from the center, as seen in Figure 1. Indeed, average commute times increased 0.9% for low-income workers, while high income workers saw a 0.5% reduction. Consistent with this finding, we estimate an increase in segregation as measured by the dissimilarity index.[62]

6.2.2 Results with Only Work Travel

We see much larger inequality effects by including travel to consume non-tradables. To see how our results change when consumption travel is not accounted for, we set the shares of non-tradable consumption for both worker groups to zero, $\alpha(\theta) = 0$, $\alpha(T, \theta) = 1 - \gamma(\theta)$,[63] and assume that all workers are employed in the tradable sector, $\lambda_{ni}(S, \theta) = 0$.[64]

Column 2 of the top panel of Table 3 presents the results from the re-estimated model. We find that the DTL improves welfare, $\bar{U}(\theta)$, for both worker groups. Welfare increases by 1.44% for high-income workers, while welfare increases by 0.84% for low-income workers. Expected income net of commuting, or access to employment, increases by 1.43% for high-income workers and by 0.43% for low-income workers. We do not capture improvements in consumption access. High income workers experience a 0.6 p.p. larger increase in welfare compared to low income workers, a three-fold underestimation relative to our baseline results.

When estimated without consumption trips, the model understates the welfare effects of the DTL because it ignores the fact that workers value access to consumption.[65] Furthermore, the

[62] Our measure of dissimilarity is $\frac{1}{2}\sum_n |R(+)/R(-) - R(+)\lambda_n(R,+)/[R(-)\lambda_n(R,-)]|$.

[63] In this model, workers make no consumption trips. Thus, this setup does not use our data on trips to consume non-tradables.

[64] Workers are employed in the same location as in the baseline. Our setup mirrors most standard quantitative spatial models (e.g., Ahlfeldt et al. 2015; Heblich, S. Redding, and D. Sturm 2020); all consumption is "tradable", no consumption requires travel, and the production of goods can happen anywhere.

[65] We find similar results for an alternative comparison simulation where we assume that all consumption is local to the neighborhood of residence (see Table A15). See Online Appendix F for details.

model also severely underestimates the inequality effects of the line in two key ways. First, the model fails to capture the disproportionate gains in access to consumption for high-income workers. Second, this version of the model misses the re-organization of non-tradable production across the city in response to changes in consumption travel, worsening access to employment for low-income workers.[66]

6.3 Model Decomposition

In Online Appendix Section F.3, we conduct a decomposition exercise to assess how robust our inequality findings are to different model assumptions. We find no qualitative differences in these new results relative to the baseline version of our model.

Our reference specification will be the version of our model with no spillovers, as in Column 1 of the bottom panel of Table 3. First, low-income workers spend less on non-tradables than the high-income. Thus, they experience smaller gains from being better able to access consumption opportunities.[67] Second, our calibration assumes low-income workers could have slightly lower travel costs than high-income workers, so they gain less from shorter travel times. Third, we assumed low-income workers have higher travel elasticities; hence, in equilibrium, they can better substitute to attractive work and consumption locations. Therefore, they may benefit less from lower travel times. Lastly, our calibration assumes low-income workers have relatively more-dispersed residential preferences; hence, they may be less able to move to take advantage of improved access elsewhere. Shutting down each channel in turn does not affect our numerical results much.

[66] What is the role of line placement in driving our results? In Online Appendix Section F.2, we simulate the removal of the North-South Line (NSL), a trunk line primarily serving more diverse neighborhoods. When the NSL is removed, welfare falls for both low and high types, by 4.2% and 2.0% respectively. Low income workers experience a 2.2 p.p. greater loss in welfare compared to high income workers. Both types experience reductions in consumption and employment access. Thus, line placement matters when considering the implications of transit expansion on welfare and equity.

[67] However, lower overall non-tradable consumption weakens the extent to which non-tradable jobs move to less attractive locations, attenuating the extent to which access to employment for low-income workers falls.

6.4 Impacts of Other Lines in Singapore

In Online Appendix Section F.2, we consider the role of line placement in driving our results. We simulate the removal of the North-South Line (NSL), a trunk line primarily serving more diverse neighborhoods. When the NSL is removed, welfare falls for both low and high types, by 4.41% and 2.08% respectively. Both types experience reductions in consumption and employment access, and inequality rises by 2.33%. Thus, line placement matters when considering the implications of transit expansion on welfare and equity. We also consider what happens if the entire rail system never existed, maintaining the current level of bus service. We predict that welfare falls by about 13-14% for low and high types respectively, driven by reductions in consumption and employment access of about 4-5% and 6-7%.

7 Conclusion

This paper demonstrates that consumption trips have important implications for the inequality effects of public transit expansions. We develop an urban spatial model with heterogeneous worker groups, low- and high-income, which incorporates both commuting and travel to consume non-tradable goods and services. We use the model to study the impact of the Downtown Line (DTL) in Singapore. We find that the DTL improves welfare for high income workers by 1.8%. Abstracting away from travel to consume non-tradables results in a three-fold underestimation of the disparate impact of the DTL across worker groups. That is, absent consumption travel, we find that both worker groups benefit, but a slightly larger share of the gains goes to high-income workers. This underestimation follows because low-income non-tradable sector jobs reorganize spatially. Aggregate welfare gains are also underestimated, ignoring gains in access to consumption opportunities for both groups. Our results sound a note of caution to a transit planner: line placement matters. Policymakers should consider the welfare and distributional implications of changes in consumption patterns, in addition to commuting, as they plan their mass transit systems.

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Figures and Tables

Table 1: Gravity Estimation

	Travel Share Conditional on Residence			
	Commuting (High Income)	Commuting (Low Income)	Consumption (High Income)	Consumption (Low Income)
	(1)	(2)	(3)	(4)
Travel Time (Minutes)	-0.051^{***}	-0.070^{***}	-0.053^{***}	-0.094^{***}
	(0.001)	(0.002)	(0.001)	(0.004)
Dest. FE	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes
IV (geog. dist)	Yes	Yes	Yes	Yes
R^2	0.854	0.660	0.792	0.576
Adjusted R^2	0.846	0.643	0.781	0.554

*p<0.1; **p<0.05; ***p<0.01

Notes: This table displays results from estimating Equation (27) in Columns 1 and 2, and Equation (31) in in Columns 3 and 4, for high- and low-income workers respectively. Destination and origin fixed effects are included, as well as an instrument, for mean travel time, of straight-line distance between neighborhood centroids. Each observation is a bilateral share of travel to a destination, conditional on residence, from 2015 fare card data. Mean travel times are computed, averaging across subzone pairs, before DTL2 opened.

Table 2: Employment Density and Idiosyncratic Productivity by Worker Type

	Parameter			
	$\varepsilon(L,+)$		$\varepsilon(L,-)$	
	(1)	(2)	(3)	(4)
Log Income (High Type)	2.075***	2.912***		
	(0.482)	(0.641)		
Log Income (Low Type)			3.637**	5.023***
			(1.410)	(1.380)
Exclude Outliers	No	Yes	No	Yes
R^2	0.328	0.378	0.149	0.280
Adjusted R^2	0.310	0.359	0.127	0.259
Residual Std. Error	0.384	0.380	0.527	0.489

*p<0.1; **p<0.05; ***p<0.01

Notes: This table displays estimation results for Equation (35) in Section 5. The dependent variable is $\varepsilon(L,\theta)$ multiplied by expected income by residence, as estimated in the model, and the explanatory variable is observed income by residence from the 2015 Singapore General Household Survey. We consider high-income workers (above the 25th percentile) in Columns 1 and 2, and consider low-income workers (below the 25th percentile) in Columns 3 and 4. In Columns 2 and 4, we drop the four neighborhoods with the highest income.

Table 3: Impact of the Downtown Line

Main Estimates	**(1) Full Model**		**(2) Only Commuting**	
% Change in	Low Type	High Type	Low Type	High Type
Welfare, $\bar{U}$	0.12	1.82	0.84	1.44
Access to Consumption, $\mathbb{S}$	1.01	1.39	0	0
Access to Employment, $\mathbb{W}$	-0.86	0.53	0.43	1.43
Gap in Welfare Impact	1.70		0.60	
Segregation	0.98		1.79	
No Spillovers	**(1) Full Model**		**(2) Only Commuting**	
% Change in	Low Type	High Type	Low Type	High Type
Welfare, $\bar{U}$	-0.17	1.69	0.55	1.32
Access to Consumption, $\mathbb{S}$	0.92	1.25	0	0
Access to Employment, $\mathbb{W}$	-1.08	0.51	0.59	1.37
Gap in Welfare Impact	1.86		0.77	
Segregation	0.86		0.59	

Notes: This table displays results from the counterfactual estimation in Section 6. The top panel displays the main estimates while the bottom panel sets agglomeration and residential spillover parameters to zero. Column 1 displays results from the full model. Column 2 shows results abstracting away from consumption trips. Welfare, access to consumption and access to employment are defined by the model in Section 4. Gap in Welfare Impact is the difference in percentage changes in welfare across high- and low-income workers. Segregation is measured by the dissimilarity index.

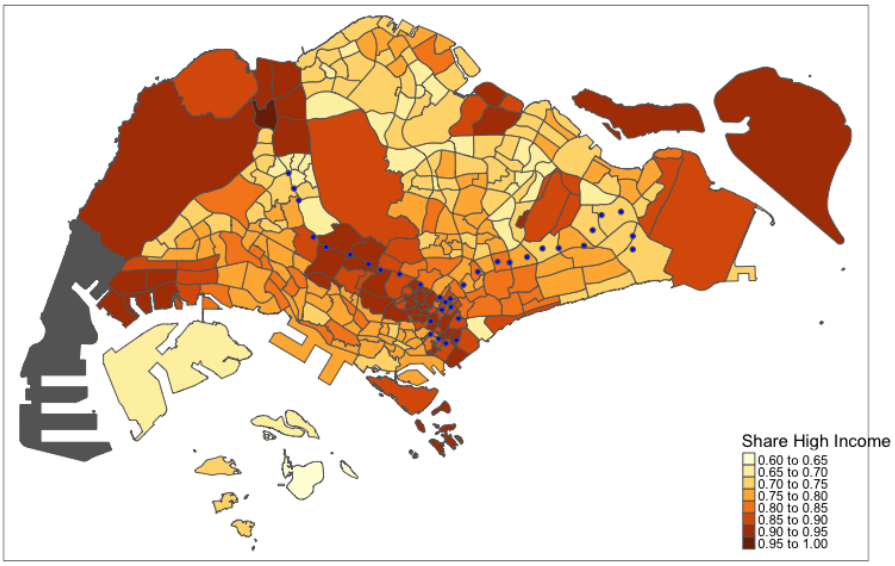
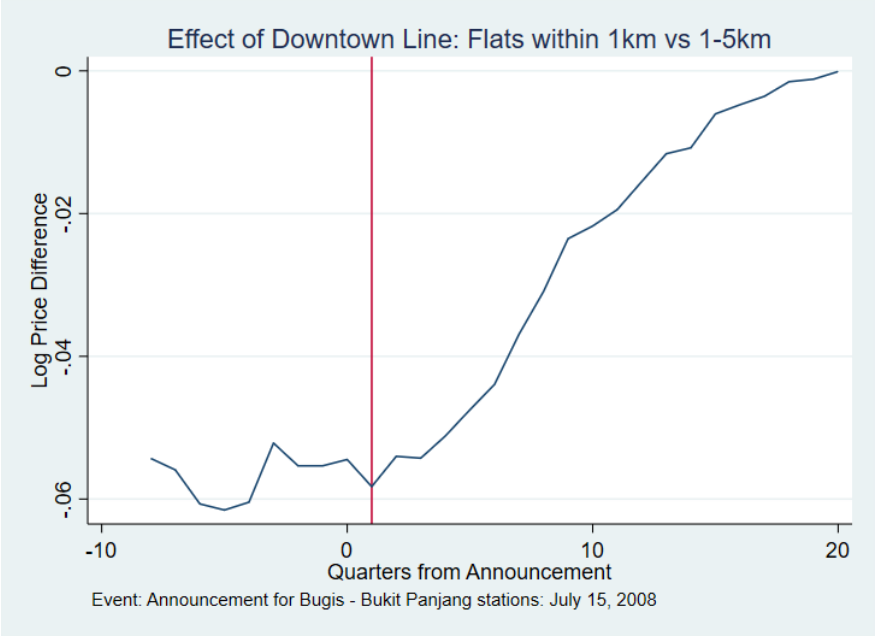
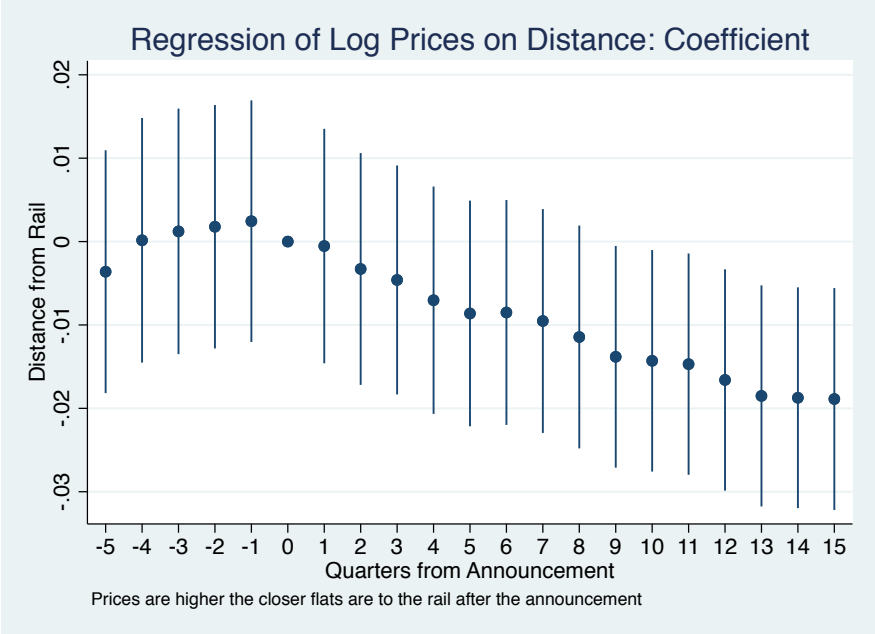


Figure 1: High-Income Residential Population Share and the Downtown Line

Notes: This figure plots the share of high-income residents in each neighborhood, or subzone, according to farecard data in 2015. Residences are imputed from the modal morning origin and modal evening destination for each Adult and Workfare farecard. High-income residents are those earning above the 25th percentile. The Downtown line stations are plotted in blue. Neighborhoods shaded in grey have population share below 0.1%.



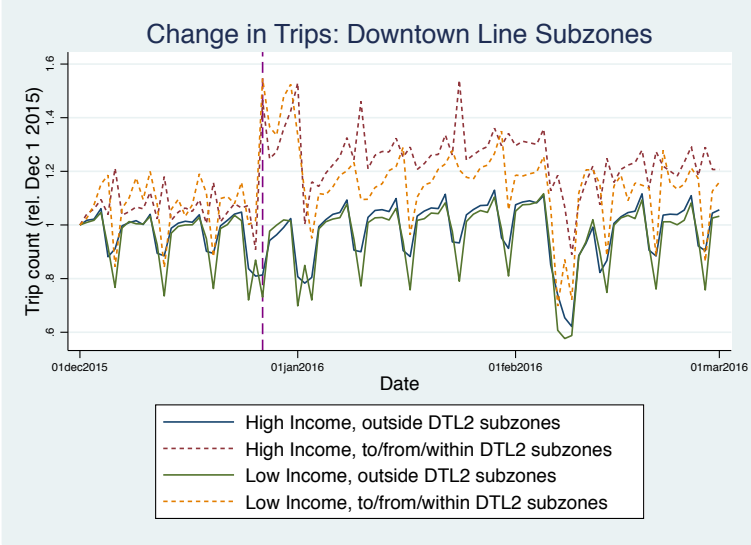
(a) Event Study of DTL2 Opening, Comparing Flats Near and Far



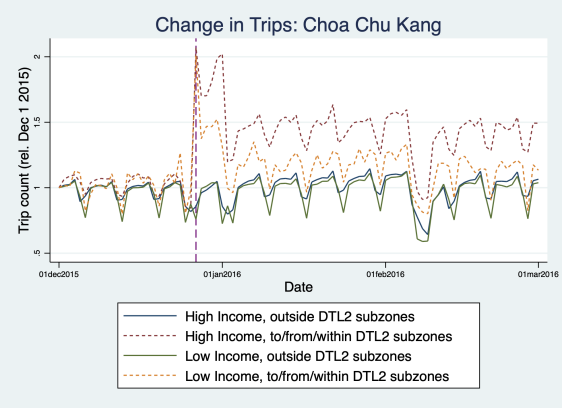
(b) Prices vs. Distance to DTL Over Time

Figure 2: Relationship between Prices and Distance to the Downtown Line Over Time

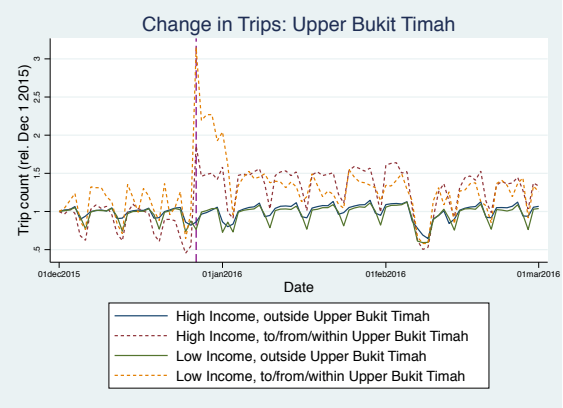
Notes: The top panel plots the log difference in mean residential prices between government apartments within 1km and between 1-5 km of a DTL2 station in each quarter, relative to the announcement date of the line alignment on 15 July 2008 (red vertical line). The bottom panel plots the relationship, over time, between log apartment prices and the distance (in km) from DTL2. We estimate Equation (42) and plot the coefficients γ from that regression. Apartment transactions are grouped by quarter, from 2007 to 2011, relative to the announcement date of the alignment of DTL2. We restrict our sample to addresses within 5km of a DTL2 station. Our data comprise the universe of resale transactions for Housing and Development Board flats in Singapore.



(a) Impact On Aggregate Travel



(b) Choa Chu Kang



(c) Upper Bukit Timah

Figure 3: Impact of the Downtown Line on Travel by High- and Low-Income Workers

Notes: This figure plots the daily volume of trips, over time, to and from: (a) subzones with and without a DTL station; (b) Choa Chu Kang; and (c) Upper Bukit Timah, compared to all other stations, by high- and low-income workers (above and below the 25th percentile). Farecard data span December 2015 to February 2016 and are from the Land Transport Authority of Singapore. The number of trips is normalized with respect to that of December 1 2015.

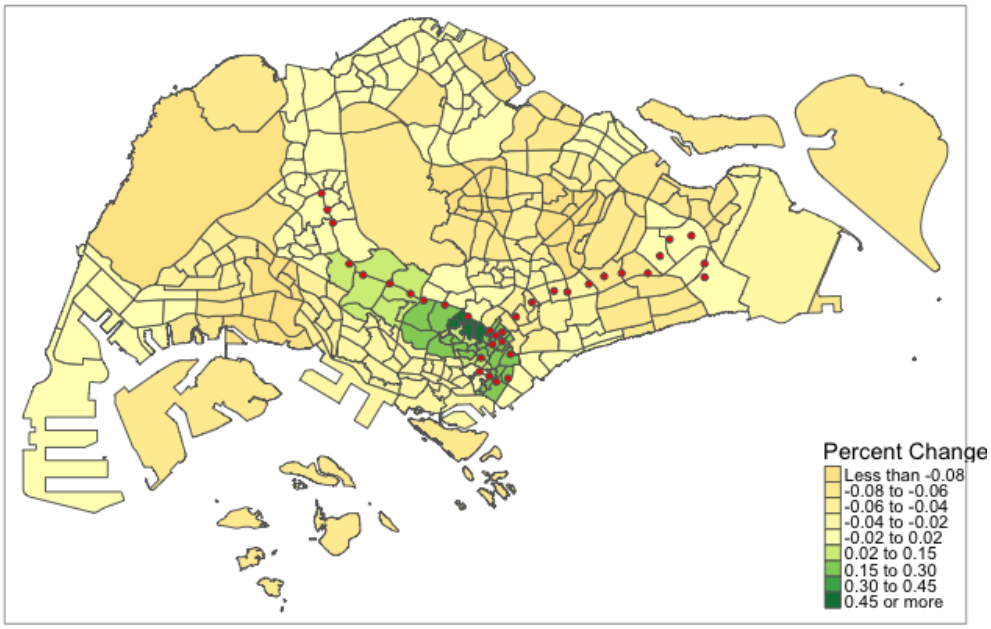


Figure 4: Model Prediction: Change in Consumption Trips

Notes: This figure displays the predicted changes in total consumption trips in the counterfactual estimation in Section 6.